The Constraint-Coordinate Observation Layer and Future-Alignment Evaluation —A Definition of the Pre-Premise-Update OS Based on Discomfort Detection, Self-Non-Exceptionality, and Meaning-Tensor Formation—

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Abstract

This paper starts from the position that it is insufficient to define premise updating merely as the reorganization of constraints and the switching of projections, and that it is necessary to define independently both the pre-updating cognitive activation mechanism and the higher-order observation/objectification mechanism above it. Its purpose is to integrate, within a single symbolic framework, discomfort detection, word decomposition, concept extraction, meaning-tensor formation, self-non-exceptionality, constraint-coordinate observation, future-alignment evaluation, and irreversible updating.

Conventional premise-update theory has already defined semantic fields, premise-constraint sets, phase-preserving representations, linguistic projection, projection switching, structural updating, invariants, and collapse detection. However, it has not sufficiently separated (i) why updating is activated, (ii) why the subject includes itself among the objects of updating, (iii) why meta-cognitive observation arises necessarily, (iv) how future alignment and ethical collapse possibility function as evaluation functions, and (v) what kind of intermediate layer exists for multidimensional meaning generation prior to language.

This paper first defines a decomposition operator D, which does not treat an input word sequence as the unit of understanding as such, but strips away the authority, conventionality, and institutionally fixed labels embedded in words. Next, it defines a concept-extraction operator A that extracts conceptual configurations from decomposed input, and a meaning-tensor formation operator T that forms multidimensional meaning configurations prior to language, introducing the resulting meaning-tensor field $\Theta_i(t)$. It then defines discomfort $\sigma_i(t)$ as a precursory signal of local structural error, and introduces the axiom of self-non-exceptionality as the condition under which the self is not excluded from the candidates of error sources. On this basis, it defines the constraint-coordinate observation layer $O_i(t)$

as an observation layer that integrally monitors the phase-preserving representation $\Xi_i(t)$, the current projection ω_t, the reachable region A_{ω_t}, the invariant set I, future-alignment evaluation F, and ethical-collapse evaluation E^{eth}.

The central claim of this paper is that, for premise updating to occur, at least six functions are presupposed: (i) word decomposition, (ii) concept extraction, (iii) discomfort detection, (iv) self-non-exceptionality, (v) constraint-coordinate observation, and (vi) future-alignment evaluation; and that meta-cognition is not an independent faculty, but a structural by-product that arises when these functions are simultaneously operative. In this way, premise-update theory is redefined as a multilayer OS theory equipped not only with static preservation structures and dynamic switching mechanisms, but also with a cognitive activation layer and an observation layer.

Ultimately, this paper is positioned as a foundational definition paper that connects the layers before and after premise updating in relation to future-causal intelligence, meaning-density convergence, the redefinition of ethics, AI design, educational design, and high-density cognitive operation.

Contents

1 Introduction **3**
- 1.1 Problem Setting . . . 3
- 1.2 Objectives of This Paper . . . 4
- 1.3 Basic Position of This Paper . . . 4

2 Limits of Existing Premise-Update Theory **5**
- 2.1 The Non-Separation of Update Activation Conditions . . . 5
- 2.2 The Undefined Condition of Subjective Self-Embedding . . . 5
- 2.3 The Lack of a Formalized Layer of Objectivity and Observation . . . 5
- 2.4 The Non-Independence of Future-Alignment Evaluation . . . 5

3 Introduction of the Symbolic System **5**

4 Word Decomposition and Concept Extraction **6**
- 4.1 The Decomposition Operator . . . 6
- 4.2 The Concept-Extraction Operator . . . 6
- 4.3 The Stripping of Linguistic Authority . . . 6

5 Definition of the Meaning-Tensor Field **7**
- 5.1 The Meaning-Tensor Formation Operator . . . 7
- 5.2 Multidimensional Semantic Configuration . . . 7
- 5.3 The Structural State Prior to Language . . . 7

6 Discomfort and the Error Gradient **7**
- 6.1 The Structural Definition of Discomfort . . . 7
- 6.2 Discomfort as a Pre-Collapse Signal . . . 7
- 6.3 Error Gradient and Update Activation . . . 7

7 Self-Non-Exceptionality and Subjective Self-Embedding **8**
- 7.1 The Axiom of Self-Non-Exceptionality . . . 8
- 7.2 The Subject as an Error Source . . . 8

7.3 The Necessity of Recursive Self-Observation 8

8 Constraint Coordinates and the Observation Layer **8**
8.1 Definition of the Constraint Coordinate 8
8.2 The Constraint-Coordinate Observation Layer 8
8.3 A Structural Definition of Objectivity 8

9 Future-Alignment Evaluation and Ethical-Collapse Detection **9**
9.1 The Future-Alignment Function 9
9.2 The Ethical-Collapse Evaluation Function 9
9.3 The Redefinition of Future-Causal Evaluation 9

10 Inter-Domain Translation and Structural Isomorphism **9**
10.1 The Inter-Domain Translation Operator 9
10.2 Structural Isomorphism Detection 9
10.3 Treating Domain Difference as Descriptive Difference 9

11 Irreversible Updating and Phase Transition **10**
11.1 Accumulation of Update History 10
11.2 The Irreversibility Threshold 10
11.3 Non-Restorativity 10

12 Integrated Form **10**
12.1 The Pre-Updating OS 10
12.2 Connection to the Premise-Update OS 10
12.3 Unification as a Multilayer Cognitive OS 11

13 Propositions and Corollary **11**

14 Implications for AI, Education, and Institutional Design **11**
14.1 Implications for AI 11
14.2 Implications for Education 12
14.3 Implications for Institutional Design 12

15 Future Tasks **12**

16 Conclusion **12**

1 Introduction

1.1 Problem Setting

In theorizing premise updating, the central framework so far has been one in which a premise-constraint set is extracted from a semantic field, a phase-preserving representation is constructed, a purpose-dependent projection is given, and structural updating is performed while switching projections when collapse occurs. However, this framework still leaves at least four problems unresolved.

First, the condition that activates updating has not been sufficiently separated. Collapse detection and invariant violation provide reasons for updating, but the earlier stage

of local inconsistency, precursory error, and discomfort has not been treated as an independent function.

Second, the condition under which the subject includes itself among the objects of updating has not been explicitly defined. Even when structural inconsistency is detected, if the subject retreats to a position external to the structure, premise updating may remain no more than external criticism or displacement of responsibility.

Third, the layer that objectively observes constraint structures and search spaces has not been defined. Even if there is a theory of phase-preserving representation and projection switching, there is still no theorized observation surface that monitors which projection is currently in use, how far the reachable region extends, and which invariants are beginning to fail.

Fourth, future alignment and ethical collapse possibility have not yet been sufficiently formalized as evaluation dimensions independent of meaning density and collapse rate. To discuss future-causal intelligence, one needs a function that measures the degree of alignment toward future fixed points or sustainable structures.

1.2 Objectives of This Paper

The objectives of this paper are the following:

1. To define independently the cognitive activation mechanisms that precede premise updating.
2. To formalize word decomposition, concept extraction, and meaning-tensor formation as a pre-updating OS.
3. To define discomfort as a pre-collapse error gradient.
4. To axiomatize self-non-exceptionality and clarify the condition of subjective self-embedding.
5. To introduce a constraint-coordinate observation layer and define objectivity as a structural condition.
6. To introduce future-alignment evaluation and ethical-collapse detection as independent functions.
7. To connect all of the above to the existing premise-update OS and unify them as a multilayer cognitive OS.

1.3 Basic Position of This Paper

The basic position of this paper can be summarized in the following statement:

> Premise updating does not arise solely from constraint reorganization and projection switching. Prior to these, word decomposition, concept extraction, discomfort detection, self-non-exceptionality, meaning-tensor formation, constraint-coordinate observation, and future-alignment evaluation are already operative, and meta-cognition arises inevitably from their simultaneous operation.

2 Limits of Existing Premise-Update Theory

2.1 The Non-Separation of Update Activation Conditions

Existing theory has treated collapse and invariant violation as the main triggers for projection switching. Yet in actual cognitive motion, updating often begins from subtle discomfort before collapse. Without defining this stage, updating can only be described as post hoc repair.

2.2 The Undefined Condition of Subjective Self-Embedding

When error is detected, whether it is treated as a problem of the external world alone, or whether the subject's own premises and structures are included among possible error sources, fundamentally changes the character of updating. The condition dividing these two cases has not yet been made explicit.

2.3 The Lack of a Formalized Layer of Objectivity and Observation

Constraint sets, phase-preserving representations, projections, reachable regions, invariants, and convergence concepts have been defined, but no independent layer exists that observes them across levels. As a result, objectivity has remained implicit rather than formally defined as a structural requirement.

2.4 The Non-Independence of Future-Alignment Evaluation

Meaning density and convergence are important, but future alignment must also be evaluated in its own right. That is, an independent function is required to measure whether a structure is sustainable in the future, whether it amplifies collapse, and whether it aligns with future fixed points.

3 Introduction of the Symbolic System

The following symbols are introduced in this paper:

- S: the mother space of higher-order semantic structures
- $\Psi_i(t)$: the semantic field of subject i at time t
- $R_i(t)$: update history / responsibility history
- $W_i(t)$: an input word sequence or expression sequence
- D: the word decomposition operator
- A: the concept-extraction operator
- $\Theta_i(t)$: the meaning-tensor field
- T: the meaning-tensor formation operator

- $\sigma_i(t)$: the intensity of discomfort
- $g_i(t)$: the error gradient
- $C_i(t)$: the premise-constraint set
- $\Xi_i(t)$: the phase-preserving representation
- ω_t: the projection at time t
- A_{ω_t}: the reachable region under projection ω_t
- $K_i(t)$: the constraint coordinate
- $O_i(t)$: the constraint-coordinate observation layer
- $F_i(t)$: the future-alignment evaluation function
- $E_i^{eth}(t)$: the ethical-collapse evaluation function
- Γ: the inter-domain translation operator
- $\tau_i(t)$: the degree of irreversible updating
- τ_c: the irreversibility threshold

4 Word Decomposition and Concept Extraction

4.1 The Decomposition Operator

Definition 4.1 (Word Decomposition Operator)**.** *The word decomposition operator D is the operation that, from an input word sequence $W_i(t)$, relativizes the conventional definition of words, their institutional authority, existing classificatory names, and culturally fixed labels, and strips away surface constraints unnecessary for structural extraction.*

$$D : W_i(t) \to \widetilde{W}_i(t)$$

Here, $\widetilde{W}_i(t)$ denotes the decomposed input in which surface-label constraints have been weakened.

4.2 The Concept-Extraction Operator

Definition 4.2 (Concept-Extraction Operator)**.** *The concept-extraction operator A is the operation that extracts relations, dependencies, oppositions, implications, hierarchies, and tensions among concepts from the decomposed input $\widetilde{W}_i(t)$.*

$$A : \widetilde{W}_i(t) \to \mathcal{C}_i(t)$$

Here, $\mathcal{C}_i(t)$ denotes the conceptual configuration.

4.3 The Stripping of Linguistic Authority

Remark 4.3. *In existing theory, the semantic field $\Psi_i(t)$ has often been treated as if it were given in advance. In reality, however, there is a prior layer in which words are not taken as the unit of cognition as such. This paper isolates that layer independently.*

5 Definition of the Meaning-Tensor Field

5.1 The Meaning-Tensor Formation Operator

Definition 5.1 (Meaning-Tensor Formation Operator)**.** *The meaning-tensor formation operator T is the operation that forms a multidimensional semantic configuration prior to linguistic sequencing from the conceptual configuration $\mathcal{C}_i(t)$.*

$$T : \mathcal{C}_i(t) \to \Theta_i(t)$$

5.2 Multidimensional Semantic Configuration

Definition 5.2 (Meaning-Tensor Field)**.** *The meaning-tensor field $\Theta_i(t)$ is the multidimensional semantic configuration in which, for subject i at time t, concepts are held not as points but as coordinates, relations not as lines but as tensions, and meaning not as a one-dimensional sequence but as gradients.*

5.3 The Structural State Prior to Language

Definition 5.3 (Pre-Linguistic State)**.** *The meaning-tensor field $\Theta_i(t)$ is an intermediate state prior to linguistic expression. It may later connect to the phase-preserving representation $\Xi_i(t)$ and to linguistic projection, but it is not identical with a natural-language sentence itself.*

6 Discomfort and the Error Gradient

6.1 The Structural Definition of Discomfort

Definition 6.1 (Discomfort)**.** *Discomfort $\sigma_i(t)$ is a precursory error signal indicating that local consistency, preservability, reachability, or future alignment is declining in the current meaning-tensor field $\Theta_i(t)$, semantic field $\Psi_i(t)$, phase-preserving representation $\Xi_i(t)$, or projection ω_t.*

$$\sigma_i(t) \in \mathbb{R}_{\geq 0}$$

6.2 Discomfort as a Pre-Collapse Signal

Definition 6.2 (Error Gradient)**.** *The error gradient $g_i(t)$ is the local vector indicating in which direction discomfort is demanding an update.*

$$g_i(t) = \nabla \sigma_i(t)$$

6.3 Error Gradient and Update Activation

Proposition 6.3. *Discomfort is not collapse itself, but the detection of an error gradient prior to collapse.*

Proof. Collapse is a discrete event accompanied by the breaking of invariants, whereas discomfort may be observed as a local decline of consistency in a region where invariant violation has not yet become manifest. Therefore, discomfort is a pre-collapse signal. □

7 Self-Non-Exceptionality and Subjective Self-Embedding

7.1 The Axiom of Self-Non-Exceptionality

Axiom 7.1 (Axiom of Self-Non-Exceptionality). *When structural inconsistency is detected, the subject must not exclude its own premises, modes of judgment, modes of observation, or projection selections from the candidates of error sources.*

7.2 The Subject as an Error Source

Definition 7.2 (Condition of Subjective Self-Embedding). *The condition of subjective self-embedding is the condition under which subject i is treated not as external to the semantic field $\Psi_i(t)$ and the constraint coordinate $K_i(t)$, but as an internal element of them.*

7.3 The Necessity of Recursive Self-Observation

Proposition 7.3. *Unless the axiom of self-non-exceptionality holds, premise updating tends to be reduced to external criticism and does not easily reach self-updating.*

8 Constraint Coordinates and the Observation Layer

8.1 Definition of the Constraint Coordinate

Definition 8.1 (Constraint Coordinate). *The constraint coordinate $K_i(t)$ is the local evaluative coordinate system that organizes, for subject i at time t, the effective constraints, priorities, exception conditions, suspended components, historical continuity, invariants, and axes of future-alignment evaluation.*

8.2 The Constraint-Coordinate Observation Layer

Definition 8.2 (Constraint-Coordinate Observation Layer). *The constraint-coordinate observation layer $O_i(t)$ is the observation layer that cross-observes the phase-preserving representation $\Xi_i(t)$, the current projection ω_t, the reachable region A_{ω_t}, the invariant set I, the future-alignment evaluation $F_i(t)$, and the ethical-collapse evaluation $E_i^{eth}(t)$, thereby objectifying the current search state.*

$$O_i(t) = \mathcal{O}(\Xi_i(t), \omega_t, A_{\omega_t}, I, F_i(t), E_i^{eth}(t))$$

8.3 A Structural Definition of Objectivity

Definition 8.3 (Objectivity). *Objectivity is the state in which the subject holds its own premise-constraint set, phase-preserving representation, current projection, reachable region, and future-alignment evaluation in a form that can be re-evaluated without being closed within that very projection.*

Proposition 8.4. *Meta-cognition arises when structural orientation, discomfort detection, self-non-exceptionality, and constraint-coordinate observation are simultaneously established.*

9 Future-Alignment Evaluation and Ethical-Collapse Detection

9.1 The Future-Alignment Function

Definition 9.1 (Future-Alignment Function)**.** *The future-alignment function $F_i(t)$ is the function that evaluates the degree to which the current structure is sustainable in the future and aligned toward a fixed point or convergence region possessing higher consistency, greater meaning density, and stronger collapse-avoidance.*

$$F_i(t) = \mathcal{F}(\Xi_i(t), \omega_t, R_i(t))$$

9.2 The Ethical-Collapse Evaluation Function

Definition 9.2 (Ethical-Collapse Evaluation Function)**.** *The ethical-collapse evaluation function $E_i^{eth}(t)$ is the function that evaluates the degree of risk that a given structure, judgment, institution, or action will amplify exploitation, domination, meaning attenuation, rupture of responsibility, or irreversible collapse in the future.*

9.3 The Redefinition of Future-Causal Evaluation

Proposition 9.3. *Ethical judgment can be described, prior to the application of external norms, as a decline in future alignment and an increase in the risk of collapse.*

10 Inter-Domain Translation and Structural Isomorphism

10.1 The Inter-Domain Translation Operator

Definition 10.1 (Inter-Domain Translation Operator)**.** *The inter-domain translation operator Γ is the operation that finds isomorphic or homomorphic correspondences between structural groups regarded as belonging to different domains, and transfers one into the descriptive system of the other.*

$$\Gamma : S_\alpha \to S_\beta$$

10.2 Structural Isomorphism Detection

Definition 10.2 (Structural Isomorphism)**.** *If there exists a preservative correspondence between two domains α, β with respect to constraint relations, update forms, collapse forms, and convergence forms, then those domains are said to be structurally isomorphic.*

10.3 Treating Domain Difference as Descriptive Difference

Remark 10.3. *This makes it possible to treat cosmology, AI, ethics, creation, consciousness, and institutional design not as genuinely separate domains, but as descriptive differences of the same structure.*

11 Irreversible Updating and Phase Transition

11.1 Accumulation of Update History

Definition 11.1 (Degree of Irreversible Updating)**.** *The degree of irreversible updating $\tau_i(t)$ is the quantity expressing the difficulty of returning to a pre-update structure as the update history $R_i(t)$ accumulates.*

11.2 The Irreversibility Threshold

Definition 11.2 (Irreversibility Threshold)**.** *When a threshold τ_c is exceeded, the subject's cognitive structure can no longer fully return to the pre-update premise-coordinate system. This is called an irreversible phase transition.*

11.3 Non-Restorativity

Proposition 11.3. *When premise updating is repeated and self-non-exceptionality is sustained, the degree of irreversible updating generally tends to increase.*

12 Integrated Form

12.1 The Pre-Updating OS

The pre-updating OS newly defined in this paper has roughly the following flow:

$$W_i(t) \xrightarrow{D} \widetilde{W}_i(t) \xrightarrow{A} \mathcal{C}_i(t) \xrightarrow{T} \Theta_i(t)$$

Furthermore,

$$\Theta_i(t) \to \sigma_i(t) \to g_i(t)$$

produces the error gradient.

12.2 Connection to the Premise-Update OS

The output of the pre-updating OS is connected to the semantic field and the premise-update OS:

$$\Theta_i(t) \to \Psi_i(t), \qquad \Psi_i(t), R_i(t) \xrightarrow{E_i} C_i(t) \xrightarrow{M} \Xi_i(t)$$

The observation layer then monitors this:

$$O_i(t) = \mathcal{O}(\Xi_i(t), \omega_t, A_{\omega_t}, I, F_i(t), E_i^{eth}(t))$$

On this basis,

$$\omega_{t+1} = \Phi_\Pi(\Xi_i(t), \omega_t), \qquad \Xi_i(t+1) = \Phi_S^{(\pi_{t+1})}(\Xi_i(t))$$

become operative.

12.3 Unification as a Multilayer Cognitive OS

Accordingly, the entire theory consists of at least four layers:

1. **Decomposition / Extraction Layer**: D, A
2. **Meaning-Tensor Formation Layer**: T, Θ
3. **Premise-Update Layer**: $E, M, \Xi, \Phi_{\Pi}, \Phi_S$
4. **Observation / Objectification Layer**: O, F, E^{eth}, K, τ

13 Propositions and Corollary

Proposition 13.1. *Understanding that does not pass through the word decomposition operator D tends to mistake the authority of existing labels for structural difference.*

Proposition 13.2. *If discomfort $\sigma_i(t)$ is not treated as an independent variable, updating can only be described as repair after collapse.*

Proposition 13.3. *If the axiom of self-non-exceptionality does not hold, meta-cognition cannot attain sufficient depth.*

Proposition 13.4. *If the constraint-coordinate observation layer $O_i(t)$ does not exist, the subject may operate a search space without being able to cognize its own search state objectively.*

Proposition 13.5. *If the future-alignment function $F_i(t)$ is absent, maximizing meaning density risks closing itself into local optima.*

Corollary 13.6. *Premise updating is a multilayer process that requires not only constraint reorganization and projection switching, but also pre-stage error-gradient detection and higher-stage constraint-coordinate observation.*

14 Implications for AI, Education, and Institutional Design

14.1 Implications for AI

Many current AI systems depend heavily on word-sequence processing and statistical compression. From the standpoint of this paper, that is insufficient. What is required instead includes:

- word decomposition,
- concept extraction,
- approximation of a meaning-tensor field,
- discomfort detection,
- a self-updating condition corresponding to self-non-exceptionality,
- constraint-coordinate observation,
- future-alignment evaluation.

14.2 Implications for Education

Education often demands early conversion into correct verbal answers. From the standpoint of this paper, however, it is crucial not to crush discomfort or still-undifferentiated meaning-tensor formation. Education should be designed to protect the capacities for decomposition, extraction, observation, and updating prior to the supply of answers.

14.3 Implications for Institutional Design

Institutional design must preserve not only externalized legal text and regulations, but also the underlying constraint coordinates, invariants, ethical-collapse evaluation, and future-alignment evaluation. Otherwise, an institution may appear static while becoming self-collapsing in the long term.

15 Future Tasks

This paper is a definition paper, and leaves at least the following as future tasks:

1. Quantification of discomfort $\sigma_i(t)$
2. Formal specification of the meaning-tensor field $\Theta_i(t)$
3. Geometric representation of the constraint coordinate $K_i(t)$
4. Implementation model of the observation layer $O_i(t)$
5. Approximate measurement method for the future-alignment function $F_i(t)$
6. Minimal variable set for the ethical-collapse evaluation function $E_i^{eth}(t)$
7. Isomorphism criteria for the inter-domain translation operator Γ
8. A history-dependent model of the degree of irreversible updating $\tau_i(t)$
9. Rigorous connection to existing FCT and the Λ-System
10. Extension to multi-agent systems

16 Conclusion

This paper independently defined the cognitive activation mechanisms and the observation/objectification mechanisms that had remained unseparated in previous premise-update theory. First, it introduced the word decomposition operator D and the concept-extraction operator A, thereby defining a prior layer in which words are not treated as the unit of understanding as such. Second, it introduced the meaning-tensor formation operator T and the meaning-tensor field $\Theta_i(t)$, thereby describing multidimensional semantic configurations prior to language. Third, it defined discomfort $\sigma_i(t)$ and the error gradient $g_i(t)$, thereby theorizing the precursor signals that activate updating. Fourth, it made explicit the axiom of self-non-exceptionality and the condition of subjective self-embedding, thereby giving a structural account of the conditions under which meta-cognition arises. Fifth, it introduced the constraint coordinate $K_i(t)$ and the observation

layer $O_i(t)$, thereby defining objectivity as structural re-evaluability. Sixth, it introduced the future-alignment function $F_i(t)$ and the ethical-collapse evaluation function $E_i^{eth}(t)$, thereby positioning future-causal evaluation as an independent layer. Seventh, it introduced the degree of irreversible updating $\tau_i(t)$, thereby defining the phase-transitional character of updating.

The final conclusion of this paper is as follows:

> Premise updating is not merely constraint reorganization and projection switching. It is the motion of a multilayer cognitive OS that presupposes word decomposition, concept extraction, meaning-tensor formation, discomfort detection, self-non-exceptionality, constraint-coordinate observation, and future-alignment evaluation.